

Overview of CubeSat

A CubeSat is a class of standardized nanosatellites developed to make space technology more accessible for educational institutions, research organizations, and student-led projects. By following predefined size and interface standards, CubeSats reduce complexity in design, integration, and launch, enabling teams to focus on learning core satellite engineering concepts.

CubeSats are primarily used for educational, experimental, and technology demonstration purposes. They offer hands-on experience in system design, subsystem integration, and documentation, making them an ideal platform for learning satellite development without the high cost and risk associated with conventional space missions.

Pocket Cube Concept

A Pocket Cube is a miniaturized form of a CubeSat that follows a smaller standard unit size. Unlike the conventional CubeSat where 1U measures 10 cm × 10 cm × 10 cm, a Pocket Cube has a 1U dimension of **5 cm × 5 cm × 5 cm**. This significantly reduced size presents unique engineering challenges, particularly in terms of power management, component placement, thermal considerations, and structural design.

Despite its small form factor, a Pocket Cube is capable of hosting essential satellite subsystems such as power, control, and communication modules. Because of these constraints, Pocket Cubes are widely used in academic projects to encourage efficient design practices and innovative solutions within limited space.

The Pocket Cube developed in this project is intended solely for **educational purposes**. The primary objective is to understand the design process, subsystem integration, and practical limitations of miniaturized satellite systems rather than for immediate deployment in orbit.

BUS and PAYLOAD

The payload mass of a satellite bus refers to the weight of the instruments and equipment that the satellite is designed to carry and deploy in space. This includes scientific instruments, communication devices, sensors, cameras, and other mission-specific hardware. Payload mass is the total weight of the functional components that the satellite carries to perform its mission. This excludes the weight of the satellite bus itself, which is the supporting structure and subsystems needed to maintain the satellite's operation in space. The payload mass is a critical parameter in satellite design because it influences the satellite's launch requirements, including the choice of launch vehicle and the fuel needed.

In a satellite, the bus system (or satellite bus) is the main support structure and subsystem framework that keeps the satellite functional and operational in space. You can think of it as the “body” and “infrastructure” of the satellite, while the payload is the “mission equipment” (like cameras, sensors, or communication antennas).

Selection of 1.5U Pocket Cube Configuration

For this project, a 1.5U Pocket Cube configuration has been selected. Given that a standard 1U Pocket Cube measures 5 cm × 5 cm × 5 cm, the 1.5U configuration provides additional internal volume along one axis while still retaining the compact nature of a Pocket Cube.

The decision to use a 1.5U Pocket Cube instead of a 1U configuration was driven primarily by size constraints of the microcontroller used in the system. The microcontroller board, along with its required peripherals and safe mounting clearances, could not be accommodated efficiently within a 1U Pocket Cube without compromising layout, accessibility, or structural integrity.

By choosing a 1.5U form factor, sufficient space is available to integrate the microcontroller while maintaining proper organization of subsystems, improving reliability, and simplifying assembly. This choice balances compactness with practicality, making it well-suited for the educational goals of the project.

Design of Pocketcube

The design of PocketCube was mechanically based on following criterias

1. Dimension
2. Material
3. Cost

PHASE I

In the 1st phase 5 cm × 5 cm × 5 cm design was tried out, with a simple side panel and base structure. The structure is made from six identical side panels that bolt together to form a rigid cube. The design is light-weighted while keeping stiffness.

Side panel design (orthographic views)

The left-side drawings show one face panel in detail.

(a) Outer dimensions

Overall panel size: 44 mm × 44 mm

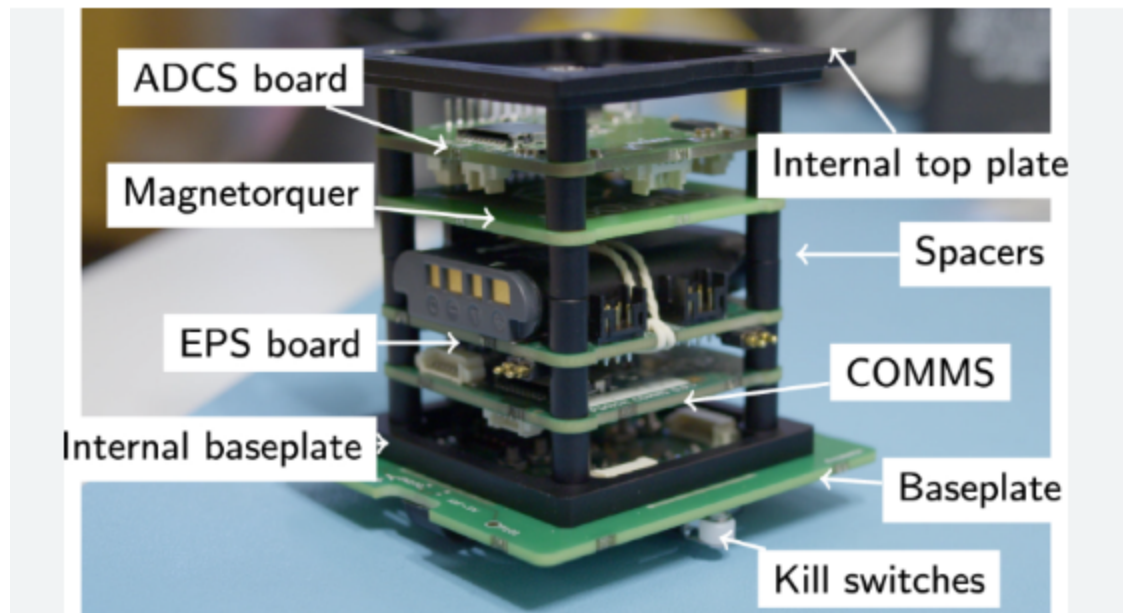
When assembled with corner edges, total cube becomes 50 mm

The missing 6 mm comes from edge frames / thickness

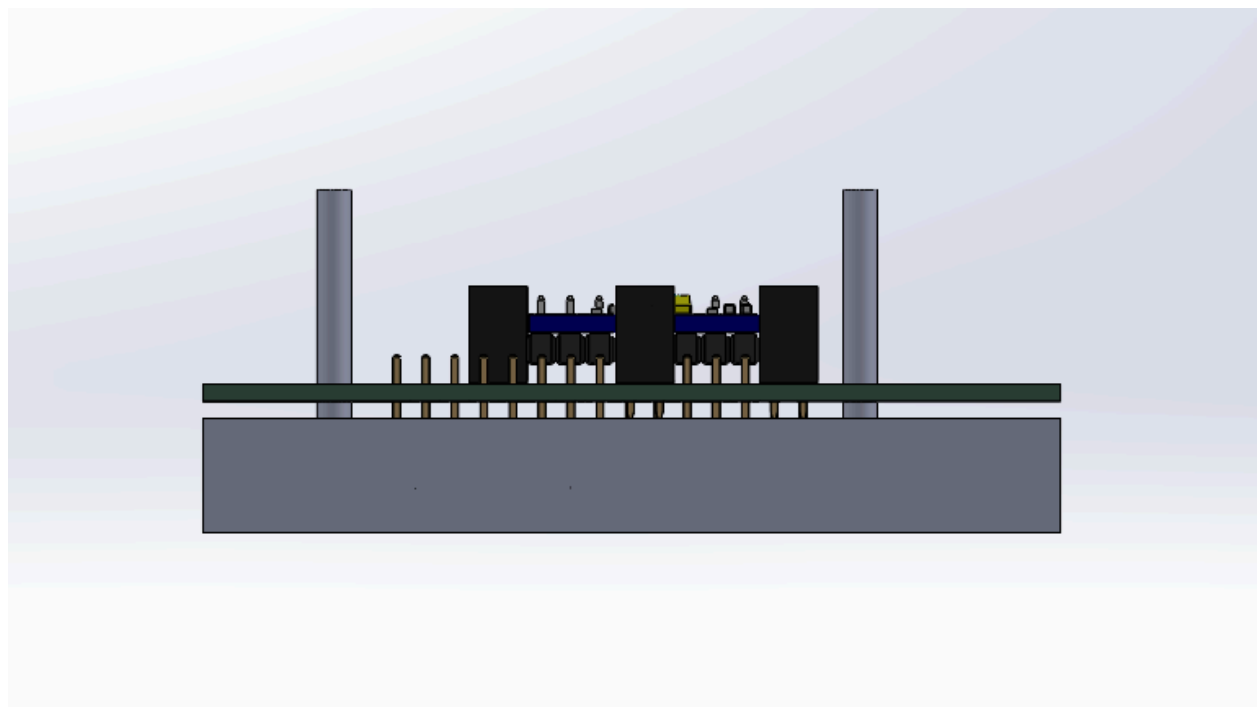
(b) Thickness

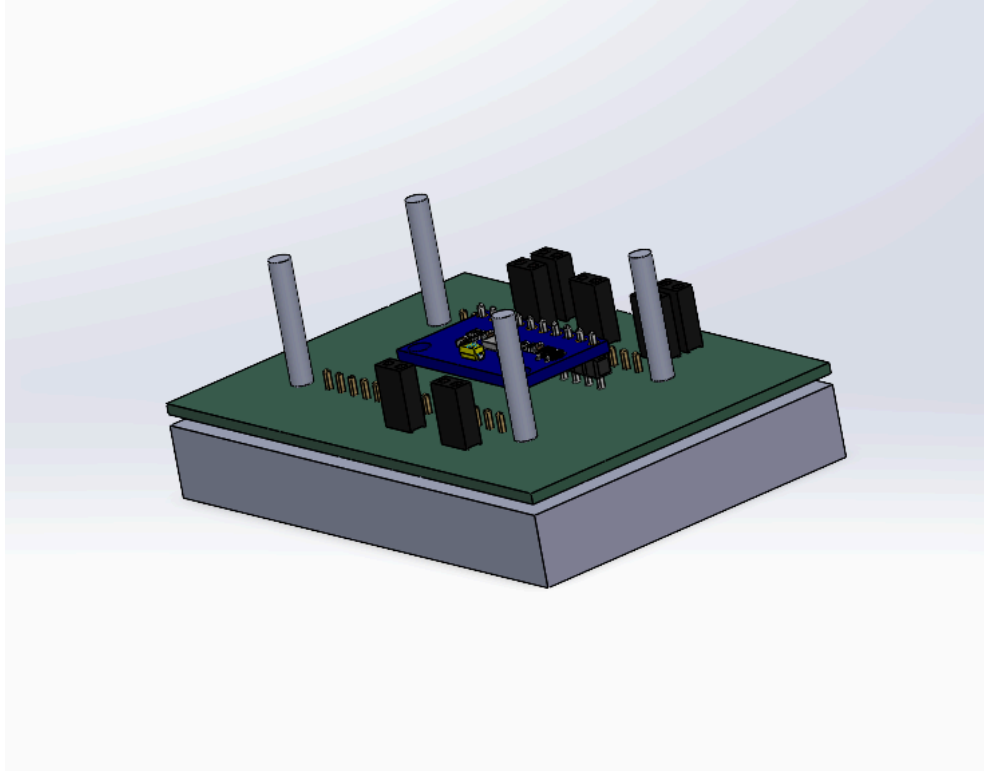
Panel thickness: 3 mm

Here, 3 mm was selected carefully to assist both strength and dimensional necessity of the device.



Mechanical design to replicate it:





Final Design :

1. Stacking of PCB
2. Space allocation for battery and other components
3. The mechanical design is centered around:
 - a. Structural frame (rails + corner members)
 - b. Side panels (PCB or aluminum): not used
 - c. Internal stack (payload + electronics)
 - d. Fastener-based assembly

The goal is structural rigidity and easy integration for educational purposes.

EXPLANATION FOR DESIGN

- a. Frames and Load bearing structure: The four vertical studs are the load bearing structure in this pocketcube. It transfers the axial and vibrational load of the structure.
- b. The top and bottom plates close the cube and clamp the rails together. It locks the frame, provides mounting points for antennas or sensors. It also act as load distribution plates to strengthen the device.
- c. The side panels are not added on this design, the side panels if added can be made of solar cells

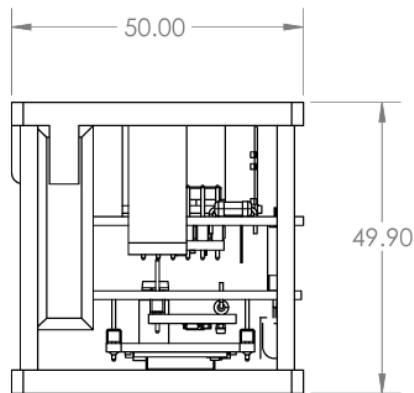
or aluminium sheets. It helps to enclose the inside of pocketcube.

- d. The electronics components are mounted on the stacked PCB configuration. Payloads are mounted on PCBs.
- e. Fasteners:

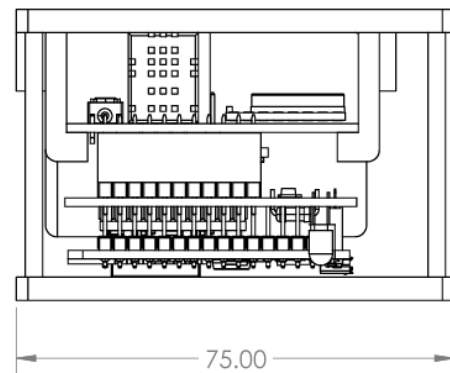
The studs are used to hold the structure. On the top face, the nuts will hold the head. The studs pass through the bottom face as well, which will have threaded holes to easily pass the shaft.

Diameter of stud= 3mm

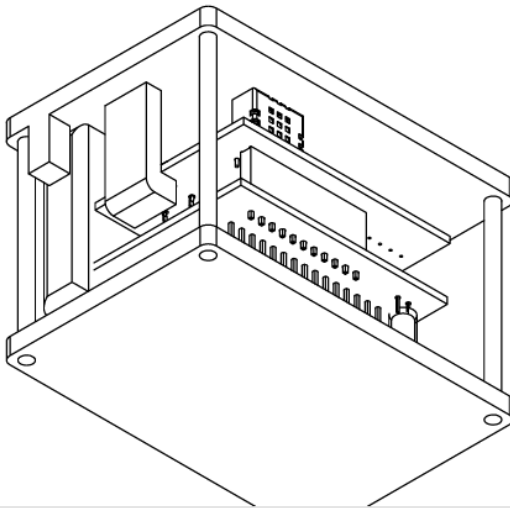
This design helps in easy assembly and disassembly allowing access to internal components.



Side View



Top View



For better understanding, please refer to the drawing pdfs provided.

CubeSat System Architecture

The architecture of a CubeSat is typically divided into multiple subsystems, each responsible for a specific function required for satellite operation. Even at the Pocket Cube scale, this modular approach is maintained to reflect real satellite design practices and to simplify development and testing.

The **On-Board Computer (OBC)** acts as the central processing unit of the CubeSat. It is responsible for executing mission logic, handling data processing, managing communication between subsystems, and controlling overall system behavior.

The **Attitude Determination and Control (ADC)** subsystem is responsible for determining the orientation of the CubeSat and applying control actions when required. This subsystem typically relies on sensor data and control logic to understand and manage the CubeSat's orientation, even though in educational models this functionality may be simplified or partially simulated.

The **Communication (COM)** subsystem enables data exchange between the CubeSat and an external ground station. It handles transmission and reception of telemetry, commands, and experimental data.

The **Electrical Power System (EPS)** is responsible for power generation, storage, regulation, and distribution. It ensures that all subsystems receive stable and reliable power within safe operating limits.

Together, these subsystems form a structured CubeSat architecture that closely resembles that of real space missions, while remaining suitable for learning and experimentation.

Microcontroller Selection

The selection of the microcontroller for the CubeSat was performed using a **decision matrix approach**. This method allowed an objective comparison of different options based on predefined criteria relevant to the project requirements.

The criteria considered during the selection process were:

- Cost
- Availability
- Physical dimensions

Each candidate microcontroller was evaluated against these criteria. Given the compact nature of the 1.5U Pocket Cube, physical size was a particularly important factor. Cost and availability were also critical to ensure feasibility within an educational setting and to allow ease of replacement or replication.

Based on the results of the decision matrix, **NodeMCU** was selected as the microcontroller for the project. It offered a suitable balance between affordability, ease of availability, and manageable dimensions, making it appropriate for integration within the Pocket Cube while supporting the architectural needs of the OBC.

Decision Marking Criteria ▾	Weight ▾	NodeMCU ▾	Esp-32Wroom ▾	STM32F103C8T ▾
Cost	10	8	7	5
Avaibility	10	10	10	10
Dimension	5	4	4	3
Total		200	190	165

Component Selection Strategy

Considering the compact size and educational nature of the 1.5U Pocket Cube, careful component selection was essential. The limited internal volume imposed strict constraints on sensor dimensions, power consumption, and integration complexity. To address these constraints while still demonstrating core CubeSat functionality, a set of essential sensing, power, and communication components was identified.

The primary focus was to include components that collectively allow environmental monitoring, motion and orientation analysis, reliable power management, and basic communication capability.

Sensor Selection

Based on project requirements, the following sensor functionalities were identified:

- Temperature
- Attitude
- Acceleration
- Speed
- Orientation

To objectively select suitable sensors, a **decision matrix** was created for each sensing requirement. The evaluation criteria primarily focused on size, power consumption, availability, cost, and compatibility with the selected microcontroller.

The following sensors were selected:

- **BMP280**: Used for temperature measurement. It offers a compact form factor, low power consumption, and reliable digital output, making it suitable for constrained Pocket Cube applications.

- **MPU-9250:** Used for attitude, acceleration, speed estimation, and orientation. This sensor integrates a gyroscope, accelerometer, and magnetometer within a single compact package, reducing component count and simplifying integration.
- **DHT Sensor:** Used for additional temperature sensing. It provides simple interfacing and low-cost environmental monitoring, which is well-suited for educational experimentation.

Power Subsystem Components

For the Electrical Power System, components were chosen to ensure sufficient energy storage, safe charging, and compatibility with small-scale solar input. A decision matrix was applied to balance capacity, size, charging efficiency, and safety.

Selected Components:

- **LiPo Battery:** Chosen for its high energy density and compact size, allowing the CubeSat to maintain operations during periods without sunlight. LiPo batteries offer reliable discharge rates for powering sensors and communication modules while keeping the overall weight low.
- **CN3065 Mini Charging Module:** Selected for its efficient charging control, overcharge protection, and compatibility with small solar panels. It ensures safe charging of the LiPo battery and protects against voltage or current fluctuations.
- **Mini Solar Panel (3–5 V):** Provides renewable energy input to the system. Its voltage output matches the CN3065 module's requirements, enabling efficient battery charging while fitting the CubeSat's compact design.

This configuration ensures a stable, safe, and efficient power subsystem that meets the CubeSat's energy requirements while adhering to size and weight constraints.

Communication Subsystem Components

For communication, a low-power long-range solution was required to demonstrate telemetry transmission and command reception.

- **LoRa RA-02 SX1278 (433 MHz):** Selected for its low power consumption, compact size, long-range communication capability, and ease of integration with the NodeMCU. Operating at 433 MHz, it is suitable for educational ground-based testing and short-range demonstrations.

The communication module was chosen through a decision matrix considering range, power usage, availability, and physical dimensions.

Detailed Schematic Description

The schematic illustrates the complete electrical design of the 1.5U Pocket Cube, showing the interconnection between the microcontroller, sensors, power subsystem, and communication module. The **NodeMCU** serves as the central control unit, functioning as the On-Board Computer and coordinating all subsystem operations.

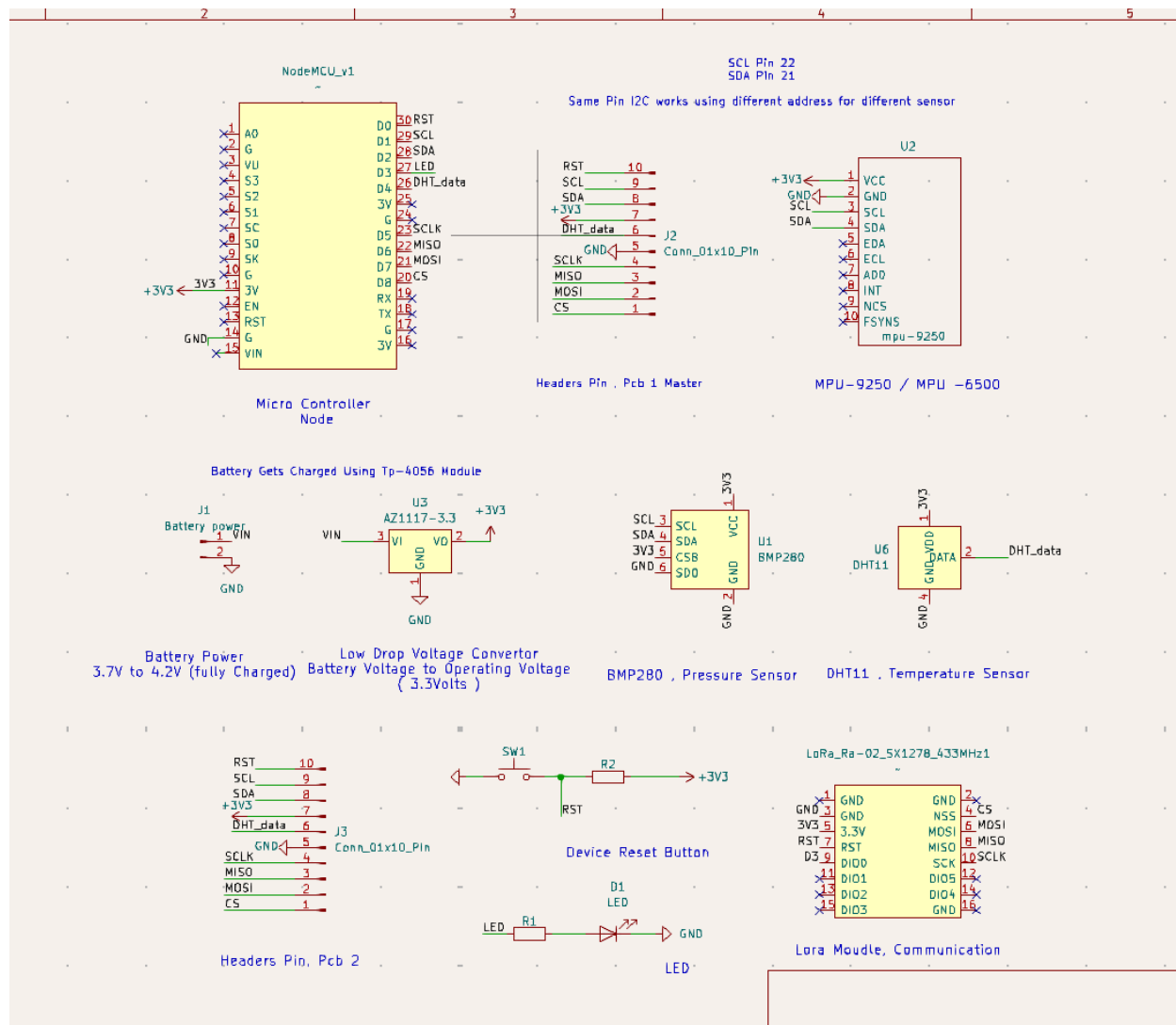
The **sensor subsystem** is primarily connected through the I²C communication bus. The BMP280 and MPU-9250 share the same I²C lines and are accessed using different device addresses, which minimizes pin usage and reduces wiring complexity. The DHT sensor uses a dedicated data line due to its single-wire communication protocol.

The **power subsystem** is based on a **LiPo battery**, selected for its high energy density and compact form factor. The LiPo battery is charged using a dedicated charging module and supplies power to the system through a voltage regulation stage. A low-dropout voltage regulator is used to convert the battery voltage to a stable 3.3 V, which is required by the microcontroller, sensors, and communication module. This ensures consistent operation and protects components from voltage fluctuations.

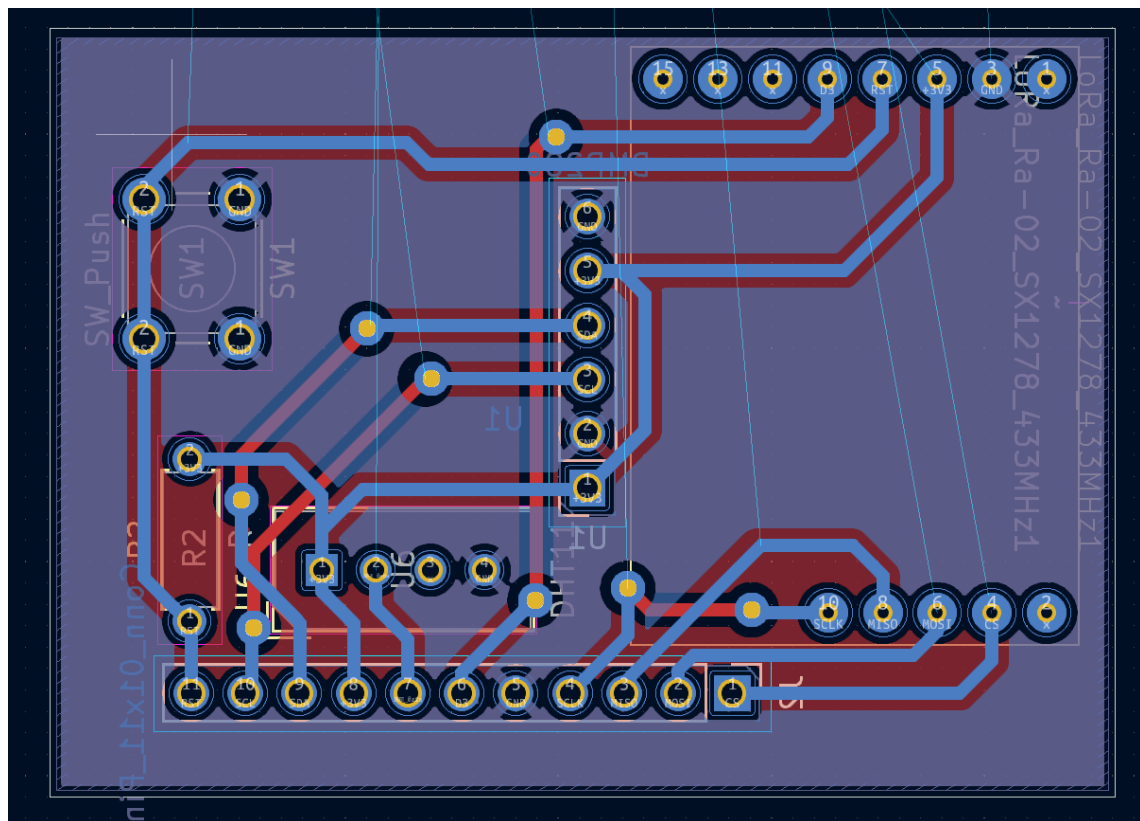
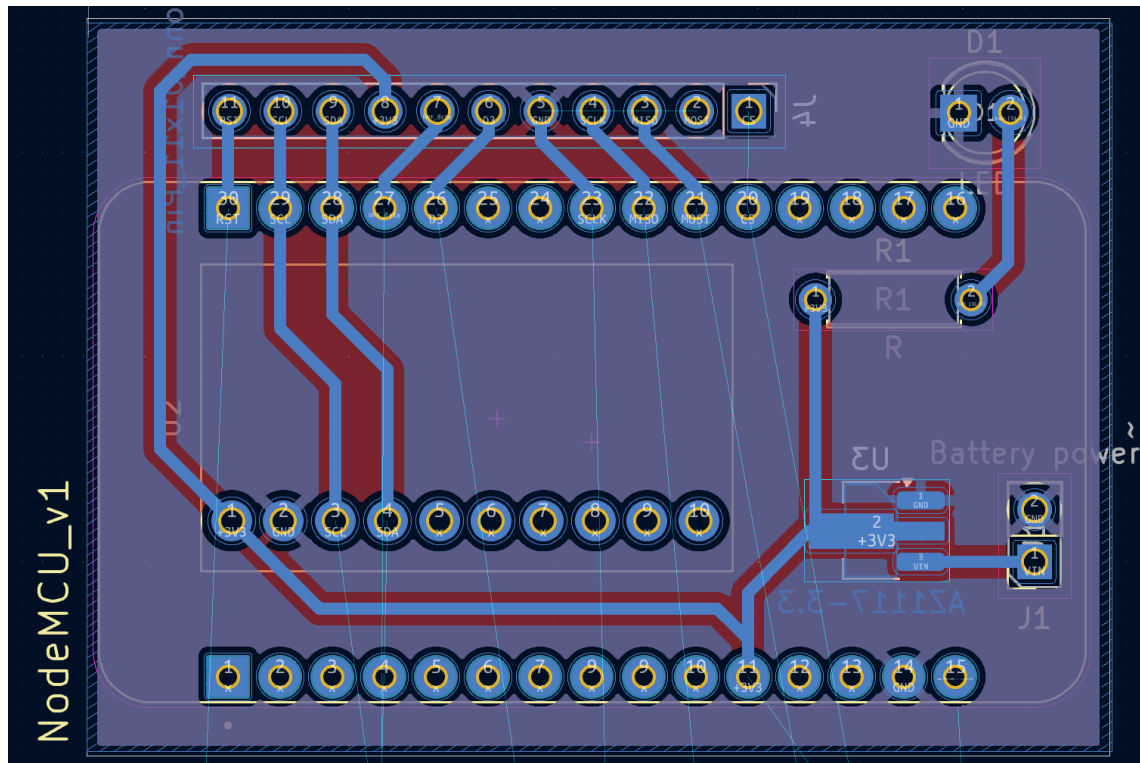
The **communication subsystem** uses the LoRa RA-02 SX1278 module, interfaced with the NodeMCU via the SPI protocol. This enables efficient long-range wireless communication for telemetry transmission and command reception.

Additional features such as programming headers, a reset switch, and status LEDs are included in the schematic to support debugging, testing, and system monitoring during development. These elements improve usability while maintaining the compact and practical design required for a Pocket Cube.

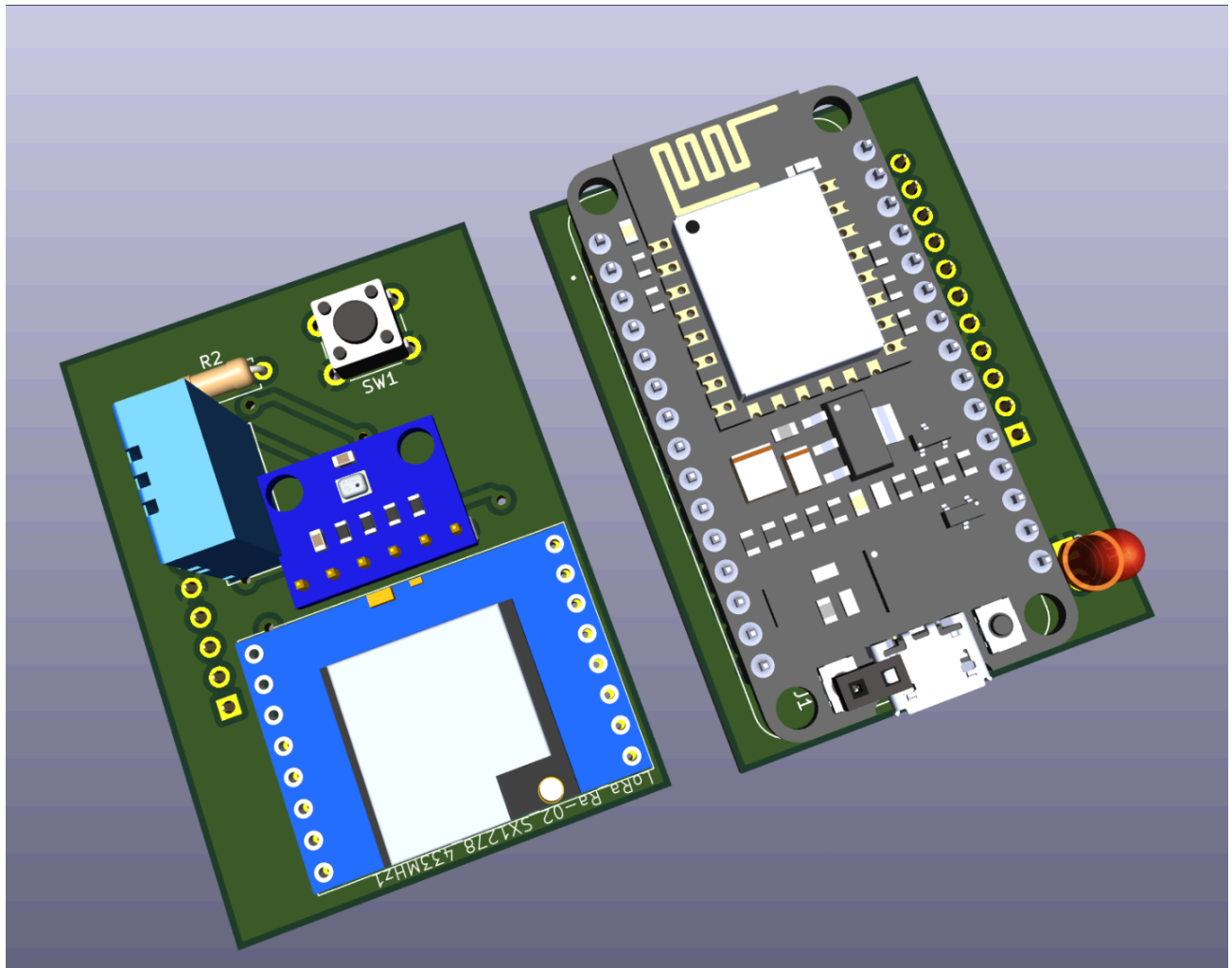
Overall, the schematic reflects a well-organized system architecture that balances functionality, power efficiency, and size constraints, in line with the educational objectives of the project.



PCB layout of Cubesat



3D view of CubeSat



Budget Estimation

S.N	Material	Cost
1.	Mpu- 9250	Rs. 750
2.	BMP280	Rs. 250
3.	Node MCU	Rs. 700
4.	Li ion 2800mah	Rs. 250
5.	CN3065 Mini	Rs. 500
6.	Mini Solar Panel	Rs. 255
7.	MQ-2 135	Rs. 368
8.	DHT	Rs. 150
9.	LoRa	Rs. 999
Total for electronics		Rs. 4,222
10.	3D printing	Weight = 50 gms(approx) 1 kg PLA= Rs. 2800 50 gms = Rs. 140
11.	Studs and nuts (3mm dia * 4)	Rs. 600
Total for mechanical		Rs. 740
TOTAL		Rs. 4962 = 5,000